

TUSHAR

Chandigarh University, Mohali, Punjab

id.tushar.01@gmail.com +91 7027009797 linkedin.com/in/tushar1631 github.com/s-tushar-01

Professional Summary

Computer Science undergraduate (AI & ML) with hands-on experience building backend systems, REST APIs, and database-driven applications using Python, C++, and PostgreSQL. Demonstrated ability to design modular, production-ready code with OOP principles and optimised SQL queries. Actively pursuing a Data Science specialisation, with completed projects in machine learning (fraud detection, climate trend analysis) and 300+ DSA problems solved across various platforms, aggregated on **Codolio**. Seeking a role where backend engineering and data skills can drive measurable impact.

Technical Skills

- **Programming Languages:** Python, C++, C, SQL
- **Backend Development:** REST API Design, CRUD Operations, Error Handling, Input Validation, Backend Architecture
- **Databases:** PostgreSQL, Schema Design, Indexing, Query Optimisation, Data Aggregation
- **Data Science / ML:** Scikit-learn, Pandas, NumPy, Matplotlib, Logistic Regression, Random Forest, Regression Models
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, System Design Fundamentals
- **Tools & Practices:** Git, GitHub, VS Code, Jupyter Notebook, Version Control, Modular Code Design, Debugging
- **Additional Exposure:** JSON, HTTP Protocol, API Integration, Linux Shell & Terminal

Education

Bachelor of Engineering – CSE (AI & ML)	2023 – 2027
Chandigarh University, Mohali, Punjab	
Intermediate (CBSE) – PCM	2022 – 2023
S.D. Sr. Sec. School, Jhajjar, Haryana — Percentage: 77.8%	
Matriculation (CBSE)	2019 – 2020
St. Francis De Sales School, Jhajjar, Haryana — Percentage: 91.8%	

Projects

Global Temperature Trend Analysis	Nov 2023 – Jan 2024
<i>Python, SQL, Pandas, NumPy, Matplotlib, Data Analysis</i>	
• Processed and cleaned large-scale climate datasets; applied regression models to identify long-term temperature patterns. • Designed SQL queries for efficient data aggregation and generated visual reports highlighting key anomalies.	
Student Record Management System (Backend)	Feb 2024 – Apr 2024
<i>C++, PostgreSQL, SQL, OOP</i>	
• Developed a backend CRUD application with PostgreSQL integration; applied OOP for modular, scalable components with validation and exception handling.	
Credit Card Fraud Detection System	Jul 2024 – Dec 2024
<i>Python, Scikit-learn, Pandas, NumPy, Matplotlib, Machine Learning</i>	
• Built an end-to-end ML pipeline using Logistic Regression and Random Forest; improved model recall to minimise false negatives on highly imbalanced datasets. • Applied preprocessing and normalisation techniques; designed scalable, OOP-based modular code architecture.	
REST API-Based Task Management Backend	Aug 2024 – Sep 2024
<i>Python, PostgreSQL, REST API, JSON, Git</i>	
• Designed and implemented RESTful CRUD APIs with optimised relational database schema (indexing) for improved query performance. • Enforced robust error handling and input validation; managed codebase via Git for structured version control.	
Online Quiz Evaluation Engine	Oct 2024 – Nov 2024
<i>C++, SQL, OOP, Data Structures</i>	
• Implemented backend logic for quiz generation, automated scoring, and timer-based execution with edge-case handling for performance optimisation.	

Certifications

- Introduction to Machine Learning – Duke University (2025)
- Structured Query Language – University of Colorado (2024)
- AI for Entrepreneurship – Intel (2024)

Achievements & Technical Exposure

- Solved 300+ DSA problems across various platforms, aggregated on **Codolio**, strengthening algorithmic thinking and problem-solving skills.
- Participated in KRAFTThink 2025 and GNA Hackathon with backend-focused problem-solving tracks.
- Hands-on experience with PostgreSQL schema design, indexing, query optimisation, and Git-based workflows.